

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Pipe Wall Thickness Calculation Report
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002
Document Revision	A01
Document Status	Issued for Approval
Originator	E. Uwaoma
Security Classification	Restricted
Issue Date	13-Dec-22

Pipe Wall Thickness Calculation Report

A01	13.12.22	Issued for Approval	E. Uwaoma	E. Nwachukwu	A. Duru	
R01	25.05.22	Issued for Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	25.05.2022	First Issue for Clients Review
A01	13.12.2022	Issued for Approval

Review Code & Status (to be filled by Client)

☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line pipe
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036	Pipeline Alignment Sheet

Table of Contents

Abbreviations	5
1. INTRODUCTION	6
1.1. Background Information	6
1.2. Purpose	7
2. REGULATIONS, CODES AND STANDARDS	7
3. PIPELINE CALCULATION BASIS	8
3.1. Location Class/Design Factor	8
3.2. Hoop Stress Design Factors	8
3.3. Road/Pipeline Crossings	8
3.4. Hydrotest Pressure.	8
3.5. Corrosion Allowance.	8
3.6. 24" Multiphase Pipeline Data	8
3.7. 36" Multiphase Pipeline Design Data	10
3.8. Pipeline Material	11
4. PIPELINE WALL THICKNESS CALCULATION	12
4.1. Wall thickness calculation procedure	12
4.2. Summary and Recommendation	12
APPENDIX 1: WALL THICKNESS CALCULATION FOR 24" API 5L GR X70 PIPE.	13
APPENDIX 2: WALL THICKNESS CALCULATION FOR 36" API 5L GR X70 PIPE.	15

Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NPS	Nominal Pipe Size
OD	Outer Diameter
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness

1.Introduction

1.1.

Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

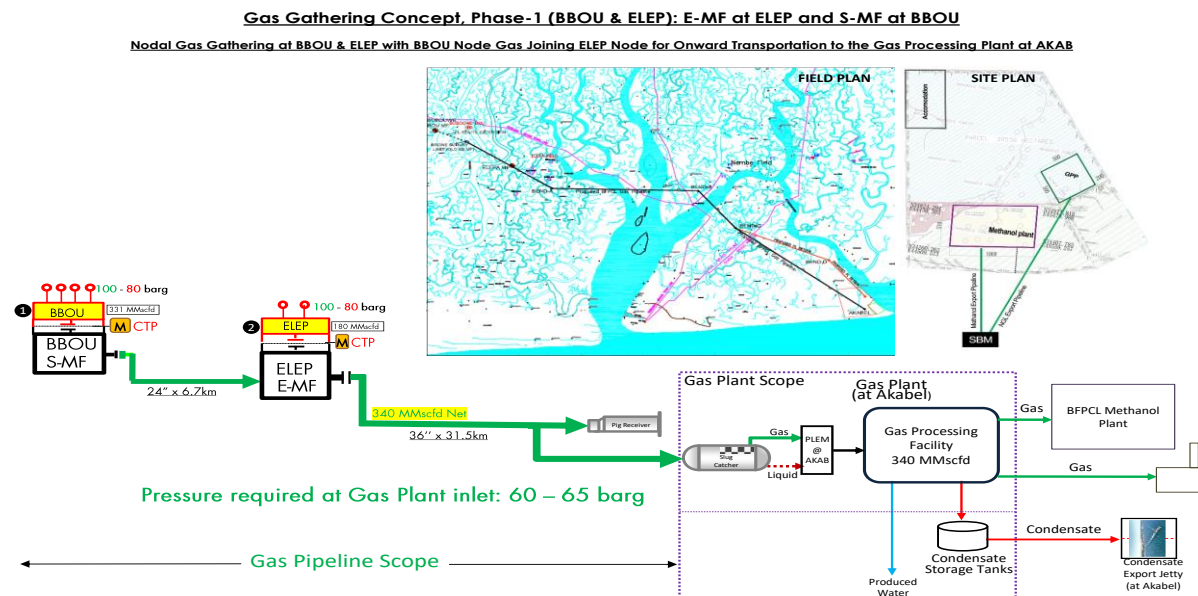


Figure 1: Gas Gathering Concept, Phase 1

1.2.

Purpose

The purpose of the document is to carry out the pipeline wall thickness design, select the corresponding wall thickness from API 5L and ASME B36.10M for the Phase 1 of this project. The design is based on material grade X70 and in accordance with industry standard codes ASME B31.8 and API 5L unless otherwise stated.

The FEED shall fully comply with all specified relevant contractual requirements.

2.Regulations, Codes And Standards

Document Number	Document Title
DEP. 31.40.20.37-Gen	Line pipe for Critical Service (February 2019)
DEP. 82.00.10.10-Gen	Project Quality Assurance
DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
ASME B31.8	Gas Transmission and Distribution Piping Systems
API 5L	Specification for Line Pipe
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Pipeline Calculation Basis

3.1. Location Class/Design Factor

Evident from the survey of the ROW the entire pipeline system is designated as location class 1. However, the wall thickness of the pipeline system shall be calculated with a design factor of 0.6 (Used for Location class 2) for the entire stretch of the Pipeline land and swamp Route. This is to mitigate against the usual occurrence in Nigeria where overtimes, new habitats spring up along pipeline ROW.

The calculation shall be carried out as per ASME B31.8 using the latest survey information to ascertain the adequacy of the selected wall thickness for the pipeline.

Where a nonstandard wall thickness is calculated after adding the corrosion allowance, The next standard wall thickness as per ASME B36.10M shall be recommended.

The selected pipe shall have a diameter to wall thickness (D/t) ratio that is not greater than 96

3.2. Hoop Stress Design Factors

The wall thickness shall be capable of withstanding the load exerted on the wall of the pipeline by the fluid internal pressure. The resulting circumferential (hoop) stress shall not exceed the specified minimum yield stress of the pipe material, multiplied by a factor applicable to the installation location. The corrosion allowance was added to the calculated wall thickness to determine the nominal wall thickness.

3.3. Road/Pipeline Crossings

A design factor of 0.6 shall be used for calculation of wall thickness for road and Pipeline crossing locations as per Table 841.1.6-2 of ASME B31.8 (2018 Edition)

3.4. Hydrotest Pressure.

The required wall thickness for hydrotest shall be $1.25 \times DP$ Per Table 841.3.2-1 of ASME B31.8 (2018 Edition)

3.5. Corrosion Allowance.

The 24" pipeline system between BBOU and ELEPA Manifold Platforms shall be designed with a corrosion allowance of 3mm while the corrosion allowance for the 36" Pipeline system between ELEPA and AKABEL shall be 6mm

3.6. 24" Multiphase Pipeline Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70(PLS2)
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	610mm/15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current
Design Life	30 years

3.7.

36" Multiphase Pipeline Design Data

The pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L – X70 (PLS2)
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	36"
Density	7,850 kg/m ³
Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	914mm/23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current
Design Life	30 years

3.8.**Pipeline Material**

The line pipe is carbon steel material grade of API-5L X70 with Product Specification Level - PSL 2 with a corrosion allowance of 6mm. The pipeline is designed for a pressure of 110 barg and the design temperature shall be 60oC.

4. Pipeline Wall Thickness Calculation

4.1.

Wall thickness calculation procedure

For determination of the required pipe wall thickness at the design pressure the following equations were used in accordance with ASME B31.8.

$$t_o = P * D / 2 * F * S * E * T \text{ Eq. 1}$$

The nominal thickness of straight section of a steel pipe is given by equation 2:

$$t_n = t_o + CA \text{ Eq. 2}$$

Where:

P = Internal design pressure (barg)

t_n = Nominal wall thickness (mm)

t_o = Pressure Containment Wall Thickness (mm)

CA = Corrosion Allowance (mm)

D = Outside diameter of pipe (mm)

S = Specified minimum yield strength of the pipe (mm HG)

E = Longitudinal weld joint factor

F = Design factor

T = Temperature derating factor

4.2.

Summary and Recommendation

The wall thickness was determined to satisfy the requirements of ASME B 31.8a Section 841.11. Based on the calculated minimum wall thickness, the recommended standard wall thickness was selected from ASME B 36.10 and as shown in the table below. The standard size selected listed below includes the corrosion

Nominal Pipe Size	OD (mm)	Design Factor	Calculated WT (mm)	Selected WT (mm)	Location Class
24"	610	0.6	14.58	15.88	1
36"	914	0.6	23.35	23.83	1

PIPELINE WALL THICKNESS DESIGN (ACCORDING TO ASME B 31.8 - 2018 Edition)				
24" NOMINAL DIAMETER (600 mm)				
<u>Location Class 1 (Designated), Design Factor 0.6:</u>				
Nominal wall thickness		$t_m = PD/(2S FET)$	(As Per 841.1.1 of ASME B31.8)	
DP	Design pressure (110 barg)	=	11	MPa
DT	Design Temperature	=	60	°C
D	Outside diameter (24")	=	610	mm
S	SMYS of pipe material- API 5L GR X70 (See Table D-1)	=	483	MPa
F	Design factor as per table 841.1.6 (For class 1- Designated)	=	0.6	
E	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)	=	1	
T	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 °C)	=	1	
t_m	Nominal wall thickness	=	11.58	mm
CA	Corrosion allowance	=	3	mm
t_c	Calculated wall thickness	$t_m + CA$	=	14.58 mm
Selected Regular standard wall thickness as per API 5L (Table 9) and AMSE B36.10M		t	=	15.88 mm
Internal diameter		D-2t	=	578.24 mm
<u>Wall Thickness check for Hydrotest</u>				
$(P_H D / 2t_c) + P_n < 90\% \text{ SMYS}$			264.14 = 54.69 %SMYS OK	
P_H	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1		13.75	Mpa
	$(P_H D / 2t)$		264.09	MPa
P_n	Hydrostatic Pressure. = $\rho g H$		0.05	MPa
D	Nominal Diameter, mm			
t	Wall Thickness, mm			
ρ	Density of water		1000	kg/m ³
g	Gravity		9.8	m/S ²
H	Height (maximum difference of elevation along the ROW) (m)		5	m

Appendix 1: Wall thickness calculation for 24" API 5L Gr X70 Pipe.

Appendix 2: Wall thickness calculation for 36" API 5L Gr X70 Pipe.

PIPELINE WALL THICKNESS DESIGN (ACCORDING TO ASME B 31.8 - 2018 Edition)				
36" NOMINAL DIAMETER (900 mm)				
<u>Location Class 1 (Designated), Design Factor 0.6:</u>				
Nominal wall thickness		$t_m = PD/(2S FET)$	(As Per 841.1.1 of ASME B31.8)	
DP -	Design pressure (110 barg)	=	11	MPa
DT -	Design Temperature	=	60	°C
D -	Outside diameter (36")	=	914	mm
S	SMYS of pipe material- API 5L GR X70 (See Table D-1)	=	483	MPa
F -	Design factor as per table 841.1.6 (For class 1-Designated)	=	0.6	
E -	Longitudinal joint factor as per table 841.17-1 (For SAW Pipe)	=	1	
T -	Temperature derating factor as per table 841.1.8-1 (For Design temp = 60 °C)	=	1	
t_m -	Nominal wall thickness	=	17.35	mm
CA -	Corrosion allowance	=	6	mm
t_c -	Calculated wall thickness	$t_m + CA$	=	23.35 mm
Selected Regular standard wall thickness as per API 5L (Table 9) and AMSE B36.10M		t	=	23.83 mm
Internal diameter		D-2t	=	866.34 mm
<u>Wall Thickness check for Hydrotest</u>				
$(P_H D / 2 t_c) + P_h < 90\% \text{ SMYS}$			263.74 = 54.60 %SMYS OK	
P_H -	Test Pressure, MPa (1.25 x DP) Per Table 841.3.2-1		13.75	Mpa
	$(P_H D / 2 t)$		263.691	MPa
P_h -	Hydrostatic Pressure. = $\rho g H$		0.05	MPa
D -	Nominal Diameter, mm			
t -	Wall Thickness, mm			
ρ -	Density of water		1000	kg/m ³
g -	Gravity		9.8	m/s ²
H -	Height (maximum difference of elevation along the ROW) (m)		5	m